

RATIONAL POWERS OF A SUM OF IDEALS

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ABSTRACT. Let R and S be two different polynomial rings over an algebraically closed field K . If I is an ideal in R , and J is an ideal in S , then it is interesting to investigate what can be said about rational powers of $(I + J)$, considered as an ideal in the fiber product of R and S . In this talk, I will define the necessary terms with examples and state the expected expression of the rational powers of $(I + J)$. I will provide classes of examples where the desired expression holds and discuss related results from our joint work with Sudipta Das, Tài Huy Hà, and Jonathan Montaño.